

Homework 4

Definitions

Define the following terms/concepts

- a.) Describe the central limit theorem and its significance to the sampling distribution for a sample proportion or mean - The Central Limit Theorem (CLT) is a fundamental concept in statistics that states that the sampling distribution of the sample mean or sample proportion approaches a normal distribution as the sample size increases, regardless of the shape of the population distribution, as long as the sample size is sufficiently large.
- b.) Margin of error - The margin of error is a measure of the uncertainty or precision associated with estimating a population parameter (such as a population mean or proportion). It quantifies the amount by which a sample estimate might deviate from the true population parameter.
- c.) Describe the “basic form” of a confidence interval - Point estimate $\pm$ Standard score $\times$ Standard error which can also be written as Point estimate $\pm$ Margin of error
- d.) Confidence level - the probability that the interval will contain the true parameter value before the data are gathered
- e.) Critical value - The point on the distribution of the test statistic that defines a set of values that call for rejecting the null hypothesis - i.e the value which defines the rejection region of a test
- f.) When are t -distribution and standard normal distribution approximately the same shape? - The t -distribution and the standard normal distribution are approximately the same shape when the sample size is large, typically considered to be greater than 30.
- g.) How are the degrees of freedom of a t -distribution related to its shape? - As the degrees of freedom increase, the t -distribution becomes more similar to the standard normal distribution, with lighter tails.
- h.) Significance level - The significance level is a predetermined threshold used in statistical hypothesis testing to determine the level of evidence required to reject the null hypothesis
- i.) Rejection region - The area in the tail (or tails) of the distribution of the test statistic for which a result is deemed statistically significant

Finding probabilities from the t and z distributions

(1) Use the appropriate table to find the probability of each z or t score.

- a.) $P(Z > 1.3) = 1 - P(Z \leq 1.3) = 0.097$
- b.) $P(Z \leq 0.33) = 0.629$
- c.) $P(Z \leq -2.1) = 0.018$
- d.) $P(t > 1.97)$ with 5 degrees of freedom $= 1 - P(t \leq 1.97) = 0.053$
- e.) $P(t \leq 2.9)$ with 2 degrees of freedom $= 0.949$
- f.) $P(t \geq -0.98)$ with 8 degrees of freedom $= 1 - P(t < -0.98) = 0.822$

Finding standard scores and α

(2) Fill in the table below with the corresponding α level and critical value for each confidence interval for a population proportion.

Confidence Level	α	critical value: $z_{1-\alpha/2}$
90%	0.1	1.645
88%	0.12	1.555
97%	0.03	2.17
96%	0.04	2.054
78%	0.22	1.227
99%	0.01	2.576

(3) Fill in the table below with the corresponding α level and critical value for each confidence interval for a population mean.

Confidence Level	Sample size	α	critical value: $t_{n-1, 1-\alpha/2}$
92%	$n = 5$	0.08	2.333
86%	$n = 8$	0.14	1.664
98%	$n = 11$	0.02	2.764
99.9%	$n = 14$	0.001	4.221
82%	$n = 40$	0.18	1.365
95%	$n = 13$	0.05	2.179

Confidence intervals

Confidence intervals for a population proportion p

(4) A survey conducted by the state of Georgia surveyed college students as part of a larger effort to characterize student behavior (Agresti and Franklin 2007). One of the questions in the survey asked students to report their political party affiliation. The table below summarizes the proportion of students who reported being Republican, Democrat, or Independent.

Political Party Affiliation	Count	$\hat{p}$
Democrat	8	0.14
Republican	36	0.61
Independent	15	0.25

Using the table above, construct a 90% confidence interval for the proportion of Independent college student voters in the state of Georgia.

```
sample.size = (8+36+15)
estimate = 15/sample.size
confidence.lvl = 0.9

CI = one.sample.prop.CI(phat = estimate,
                        n = sample.size,
                        alpha = 1-confidence.lvl,
                        verbose = T)
```

```
## [1] -----
## [1] Estimate = 0.2542
## [1] Critical Value = 1.6449
## [1] Estimated Standard Error = 0.0567
## [1] Margin of error = 0.0932
## [1] 90 % CI = [0.161,0.3475]
## [1] -----
```

(5) As part of an effort to improve health services, a health club in the United Kingdom asked club members to complete an online survey (de Vries 2023). One of the survey questions asked customers to rate their satisfaction level on a likert scale. The results of this survey question are summarized below

Table 1: Customer responses for the survey question "How satisfied are you with your membership of the organisation overall?"

Response	Count	$\hat{p}$
Completely dissatisfied	0	0.00
Very dissatisfied	0	0.00
Dissatisfied	3	0.01
Neutral	20	0.09
Satisfied	64	0.30
Very satisfied	109	0.51
Completely satisfied	19	0.09

Using the table above, construct a 95% confidence interval for the proportion of customers that are “very satisfied” or more.

```
sample.size = (3+20+64+109+19)
very.satisfied = 109
completely.satisfied = 19
estimate = (very.satisfied+completely.satisfied)/sample.size
confidence.lvl = 0.95

CI = one.sample.prop.CI(phat = estimate,
                        n = sample.size,
                        alpha = 1-confidence.lvl,
                        verbose = T)
```

```
## [1] -----
## [1] Estimate = 0.5953
## [1] Critical Value = 1.96
## [1] Estimated Standard Error = 0.0335
## [1] Margin of error = 0.0656
## [1] 95 % CI = [0.5297,0.661]
## [1] -----
```

Confidence intervals for a population mean μ

(6) The following table summarizes variables from a random sample of 25 observations from a dataset concerning housing in California taken from the 1990 census. The full dataset is featured in Aurélien Géron’s book ‘Hands-On Machine learning with Scikit-Learn and TensorFlow’ (Géron 2022). Use the table to answer parts (a) and (b)

Table 2: Summary statistics for housing data in California from the 1990 census. Estimates are derived from a random sample of 25 observations from Aurélien Geron's dataset featured in 'Hands-On Machine learning with Scikit-Learn and TensorFlow', originally comprising observations from over 20,000 districts in California

Variable	Sample Mean	Sample Standard Deviation
Median Age of Houses (per block)	24.28	13.56
Total Rooms (per block)	3263.72	2105.99
Total Bedrooms (per block)	640.52	380.55
Population of District	1861.68	1193.30
Number of Households (per block)	584.24	330.38
Median Family Income (in tens of thousands of USD)	3.74	1.54
Median House value (USD)	179320.04	109450.56

- a.) Construct and interpret a 95% confidence interval for the average median family income for households in California in 1990

```
sample.size = 25
sample.mean = round(mean(hss$median_income), 2)*10000 #converting estimate to USD
sample.sd = round(sd(hss$median_income), 2)*10000 #converting standard deviation to USD
confidence.lvl = 0.95

CI = one.sample.t.CI(xbar = sample.mean,
                     s = sample.sd,
                     n = sample.size,
                     alpha = 1 - confidence.lvl,
                     verbose = T)

## [1] -----
## [1] Estimate = 37400
## [1] Critical Value = 2.0639
## [1] Estimated Standard Error = 3080
## [1] Margin of error = 6356.8076
## [1] 95 % CI = [31043.1924,43756.8076]
## [1] -----
```

At the 95% confidence level, we estimate the average median family income in California in 1990 to be no less than \$31,043 USD and no more than \$43,757 USD. If we repeated this sampling procedure many times, 95% of the intervals constructed this way would contain the true average.

- b.) Construct and interpret a 99% confidence interval for the average number of bedrooms per block in California in 1990

```
sample.size = 25
sample.mean = round(mean(hss$total_bedrooms), 2)
sample.sd = round(sd(hss$total_bedrooms), 2)
confidence.lvl = 0.99

CI = one.sample.t.CI(xbar = sample.mean,
                     s = sample.sd,
                     n = sample.size,
                     alpha = 1 - confidence.lvl,
                     verbose = T)

## [1] -----
## [1] Estimate = 640.52
```

```
## [1] Critical Value = 2.7969
## [1] Estimated Standard Error = 76.11
## [1] Margin of error = 212.8751
## [1] 99 % CI = [427.6449,853.3951]
## [1] -----
```

At the 99% confidence level, we estimate the average number of bedrooms per block in California in 1990 to be no less than 427.6 and no more than 853.4. If we repeated this sampling procedure many times, 99% of the intervals constructed this way would contain the true average.

Choosing a minimum sample size

(7) According to a 2023 study published by the Saudi Heart Association, a survey of college students from Taibah University in Medina, Saudi Arabia, found that 24% of students reported using e-cigarettes (Alzahrani et al. 2023). Assuming a comparable rate of e-cigarette use in the U.S., what sample size is required to estimate the proportion of college students who use e-cigarettes at the 95% confidence level and within a margin of error of 1.5%?

```
m = 0.015 #desired margin of error
p = 0.24 #expected proportion of e-cigarette use based on Saudi Study
confidence.lvl = 0.95
alpha = 1 - confidence.lvl

z = qnorm(1 - alpha/2) #standard score for confidence level of 95%

n = ceiling((z^2*p*(1-p))/m^2) # compute sample size - always round UP

# a sample size of approximately
print(paste0('n = ', n, ' observations'))
```

```
## [1] "n = 3115 observations"
```

```
#are needed to achieve the desired confidence level and margin of error
```

(8) Recent concerns over the rise in global infertility have drawn significant attention to the harmful effects of microplastics, PFAS, and highly processed foods (Zhang et al. 2022; Agarwal et al. 2015). A study from 2002 published in Fertility and Sterility found that approximately 15% of couples worldwide experienced infertility (Sharlip et al. 2002). Assuming a similar rate of infertility in the U.S, compute the sample size needed to estimate infertility of U.S couples at the 99% confidence level and within a margin of error of 4%.

```
m = 0.04 #desired margin of error
p = 0.15 #expected proportion of infertile couples based on previous studies
confidence.lvl = 0.99
alpha = 1 - confidence.lvl

z = qnorm(1 - alpha/2) #standard score for confidence level of 99%

n = ceiling((z^2*p*(1-p))/m^2) # compute sample size - always round UP

# a sample size of approximately
print(paste0('n = ', n, ' observations'))
```

```
## [1] "n = 529 observations"
```

```
#are needed to achieve the desired confidence level and margin of error
```

Assumptions of estimators

(9) Describe the assumptions associated with estimating p with $\hat{p}$

1. **Random Sampling:** The sample must be selected randomly from the population of interest. Random sampling ensures that each member of the population has an equal chance of being included in the sample, allowing for unbiased estimation of the population proportion.
2. **Independence:** The observations in the sample must be independent of each other. Independence means that the occurrence or value of one observation does not influence the occurrence or value of another observation in the sample.
3. **Large Sample Size (for Asymptotic Normality):** For certain statistical methods, such as constructing confidence intervals or conducting hypothesis tests, the sample size should be sufficiently large. While there is no fixed threshold for what constitutes “large enough,” a common rule of thumb is that $np \geq 15$ and $n(1 - p) \geq 15$ to ensure the validity of results.

(10) Describe the assumptions associated with estimating μ with $\bar{x}$

1. **Random Sampling:** The sample must be selected randomly from the population of interest. Random sampling ensures that each member of the population has an equal chance of being included in the sample, allowing for unbiased estimation of the population proportion.
2. **Independence:** The observations in the sample must be independent of each other. Independence means that the occurrence or value of one observation does not influence the occurrence or value of another observation in the sample.
3. **Normality of Population Distribution (for Small Sample Sizes):** For small sample sizes (typically less than 30), it is assumed that the population from which the sample is drawn follows a normal distribution. This assumption is necessary for the validity of certain statistical inference methods, such as constructing confidence intervals or conducting hypothesis tests. However, the Central Limit Theorem suggests that for larger sample sizes, the distribution of the sample mean approaches normality regardless of the shape of the population distribution.

Significance testing

Stating hypotheses

(11) A pharmaceutical company produces a generic version of the pain reliever ibuprofen, marketing a tablet with a 200 milligram dose. Concerned about the accuracy of the dosing process, the manufacturer suspects that the machine filling the tablets may be malfunctioning leading to a smaller dose in each tablet. The manufacturer wishes to conduct a significance test to determine if the dose is significantly lower than 200 milligrams. State the null and alternative hypotheses for this test

The null hypothesis is that the mean dose of a tablet is 200 mg

$$H_0 : \mu_0 = 200$$

The alternative hypothesis is that the mean dose of a tablet is less than 200 mg

$$H_A : \mu < \mu_0$$

(12) A financial institution invests in a pharmaceutical company, which produces a widely prescribed antidepressant medication with a market value of \$100 per share. Worried about potential discrepancies in the market valuation, the institution suspects that recent market trends may have led to an underestimation of the stock's value. The institution plans to conduct a significance test to determine if the stock's value has significantly increased. State the null and alternative hypotheses for this test.

The null hypothesis is that the mean share value is \$100

$$H_0 : \mu_0 = 100$$

The alternative hypothesis is that the mean share value is greater than \$100

$$H_A : \mu > \mu_0$$

(13) A semiconductor manufacturing company produces microchips with a target defect rate of 0.1% per batch. Concerned about the quality control process, the company suspects that recent changes in manufacturing procedures may have resulted in a change in the defect rate. The company intends to conduct a significance test to determine if the defect rate is significantly different than the target rate of 0.1%. State the null and alternative hypotheses for this test.

The null hypothesis is that the proportion of defective semi-conductors is 0.001

$$H_0 : p_0 = 0.001$$

The alternative hypothesis is that that the proportion of defective semi-conductors is not equal to 0.001

$$H_A : p \neq p_0$$

The Decision Rule

(14) For each set of hypotheses, significance level, and p -value. State whether the test rejects or fails to reject the null hypothesis.

- a.) $H_0 : \mu = 0$; $H_A : \mu \neq 0$; $\alpha = 0.01$; p -value = 0.0098 Decision: reject H_0 because p -value $< \alpha$
- b.) $H_0 : p = 0.5$; $H_A : p > 0.5$; $\alpha = 0.05$; p -value = 0.086 Decision: fail to reject H_0 because p -value $> \alpha$
- c.) $H_0 : \mu = 100$; $H_A : \mu < 100$; $\alpha = 0.001$; p -value = 0.0015 Decision: fail to reject H_0 because p -value $> \alpha$
- d.) $H_0 : p = 0.9$; $H_A : p \neq 0.9$; $\alpha = 0.1$; p -value = 0.053 Decision: reject H_0 because p -value $< \alpha$

(15) For each set of hypotheses, significance level, sample size, and test statistic. Give the critical value and state whether the test rejects or fails to reject the null hypothesis.

- a.) $H_0 : \mu = 15$; $H_A : \mu > 15$; $n = 33$; $\alpha = 0.05$; $t_{obs} = 2.13$ **critical value:** $t_{32,0.95} = 1.69$; **Decision:** reject H_0 because $t_{obs} > 1.69$
- b.) $H_0 : \mu = 0$; $H_A : \mu \neq 0$; $n = 14$; $\alpha = 0.01$; $t_{obs} = -4.1$ **critical value:** $t_{13,0.995} = 3.01$; **Decision:** reject H_0 because $|t_{obs}| > 3.01$
- c.) $H_0 : p = 0.5$; $H_A : p < 0.5$; $n = 120$; $\alpha = 0.1$; $Z_{obs} = -1.5$ **critical value:** $Z_{0.1} = -1.28$; **Decision:** reject H_0 because $Z_{obs} < -1.28$
- d.) $H_0 : p = 0.05$; $H_A : p \neq 0.05$; $n = 60$; $\alpha = 0.03$; $Z_{obs} = 1.95$ **critical value:** $Z_{0.985} = 2.17$; **Decision:** fail to reject H_0 because $|Z_{obs}|$ is not greater than 2.17

Hypothesis tests for a population proportion p

(16) A toy manufacturer claims that 50% of their toy robots are defect-free. However, a quality control inspector suspects that the actual proportion of defect-free robots is different from what the manufacturer claims. To investigate, the inspector randomly selects 100 toy robots from a production batch and examines them for defects. After the inspection, the inspector finds that 40 out of the 100 toy robots are defect-free. To test their suspicion, they set up a two-tailed hypothesis test with a significance level $\alpha = 0.1$ and hypotheses $H_0 : p_0 = 0.5$ (manufacturer's claim) $H_A : p \neq p_0$.

- a.) Using the significance level above, is there enough evidence to conclude that the proportion of defect-free toy robots is different from the manufacturer's claim?

```

test.type = 'two.tail'
null.value = 0.5
sample.size = 100
estimated.proportion = 40/100
significance.lvl = 0.1

one.sample.prop.test(p0 = null.value,
                     phat = estimated.proportion,
                     n = sample.size,
                     alpha = significance.lvl,
                     test = test.type,
                     verbose = T)

## [1] ===== test results =====
## [1] test type = two.tail
## [1] H0: p0 = 0.5
## [1] HA: p != 0.5
## [1] estimate = 0.4
## [1] Estimated Standard Error = 0.05
## [1] Critical Value = 1.6449
## [1] 90% CI = [0.3194,0.4806]
## [1] Test statistic = -2
## [1] Pvalue = 0.0455
## [1] Decision: reject H0
## [1] =====

```

The test statistic is $Z_{obs} = \frac{0.40-0.50}{\sqrt{0.5(1-0.5)/100}} = -2.00$ with a two-tailed p -value of 0.0455. Since $0.0455 < \alpha = 0.1$ (equivalently, $|Z_{obs}| = 2.00 > 1.645$), at the 10% significance level we reject the null hypothesis and conclude that the proportion of defect-free toy robots is significantly different from the manufacturer's claim of 0.5

(17) A pharmaceutical company has developed a new drug designed to cure a specific bacterial infection. They claim that their drug is effective, with a cure rate of 20%. However, a group of independent researchers believes that the cure rate is less than what the pharmaceutical company claims. To investigate, the researchers conduct a clinical trial on 300 patients suffering from the bacterial infection. After the trial, they find that 70 out of the 300 patients were cured using the new drug. To test their suspicion, they set up a hypothesis test with a significance level of $\alpha = 0.05$ with $H_0 : p_0 = 0.2$, $H_A : p < p_0$

```

test.type = 'lower.tail'
null.value = 0.2
sample.size = 300
estimated.proportion = 70/300
significance.lvl = 0.05

one.sample.prop.test(p0 = null.value,
                     phat = estimated.proportion,
                     n = sample.size,
                     alpha = significance.lvl,
                     test = test.type,
                     verbose = T)

## [1] ===== test results =====
## [1] test type = lower.tail
## [1] H0: p0 = 0.2
## [1] HA: p < 0.2
## [1] estimate = 0.2333

```

```
## [1] Estimated Standard Error = 0.0231
## [1] Critical Value = -1.6449
## [1] 95% CI = [0.1855,0.2812]
## [1] Test statistic = 1.4434
## [1] Pvalue = 0.9255
## [1] Decision: fail to reject H0
## [1] =====
```

- a.) Compute the critical value for this test

the critical value is $Z_{\alpha} = -1.64$

- b.) Compute the test statistic Z_{obs}

the test statistic is

$$Z_{obs} = \frac{0.233 - 0.2}{\sqrt{\frac{0.2(1-0.2)}{300}}} = 1.44$$

- c.) Compute the p -value

the p -value is

$$P(Z < Z_{obs} | H_0 \text{ True}) = 0.9255$$

- d.) Is there enough evidence to conclude that the cure rate of the new drug is less than the rate claimed by the pharmaceutical company?

At the 5% significance level, we fail to reject the null hypothesis and conclude that the proportion of patients cured by the drug is not significantly less than 0.2

Hypothesis tests for a population mean μ

(18) A soft drink company claims that their new “Zero Sugar” soda contains zero grams of added sugar per 12-ounce can. To test this claim, a random sample of 25 cans of this soda is selected. The sample mean is found to be 1.3 grams of added sugar per can, with a standard deviation of 0.5 grams. To determine if there is enough evidence to reject the company’s claim and conclude that the soda does not contain zero grams of added sugar per can, a group of researchers is interested in testing the following hypotheses: $H_0 : \mu_0 = 0$, $H_A : \mu \neq \mu_0$ at the $\alpha = 0.01$ significance level.

- a.) Using the significance level above, is there evidence that the soda produced by the soft drink company contains added sugars?

#The null hypothesis is H_0: m0 = 0 (no added sugar)

*#The alternative hypothesis is H_A: m > m0 - this is the best option since it does not
#make since check for less than 0 added sugar.*

```
test.type = 'upper.tail'
null.value = 0
sample.mean = 1.3
sample.sd = 0.5
sample.size = 25
significance.lvl = 0.01

one.sample.t.test(m0 = null.value,
                  xbar = sample.mean,
                  s = sample.sd,
                  n = sample.size,
                  alpha = significance.lvl,
```

```
test = test.type,
verbose = T)
```

```
## [1] ===== test results =====
## [1] test type = upper.tail
## [1] H0: m0 = 0
## [1] HA: m > 0
## [1] Estimate = 1.3
## [1] Estimated Standard Error = 0.1
## [1] Critical Value = 2.4922
## [1] 99% CI = [1.0203,1.5797]
## [1] Test statistic = 13
## [1] Degrees of Freedom = 24
## [1] Pvalue = 0
## [1] Decision: reject H0
## [1] =====
```

At the 1% significance level, we reject the null hypothesis and conclude that the soda produced by the company contains significant amounts of added sugars

(19) A group of botanists is studying the growth rate of Venus fly traps in a controlled greenhouse environment. Based on historical data, the typical growth rate of Venus fly traps is believed to be 100 millimeters per year. The botanists suspect that the current growth rate in their greenhouse is higher than this historical value. To investigate, they take a random sample of 40 Venus fly traps and measure their growth rates over a year. After the study, they find that the sample mean growth rate for the 40 Venus fly traps is 105 millimeters per year with a standard deviation of 12mm. To test their suspicion, they set up a hypothesis test with a significance level of $\alpha = 0.05$

```
test.type = 'upper.tail'
null.value = 100
sample.mean = 105
sample.sd = 12
sample.size = 40
significance.lvl = 0.05

one.sample.t.test(m0 = null.value,
                  xbar = sample.mean,
                  s = sample.sd,
                  n = sample.size,
                  alpha = significance.lvl,
                  test = test.type,
                  verbose = T)
```

```
## [1] ===== test results =====
## [1] test type = upper.tail
## [1] H0: m0 = 100
## [1] HA: m > 100
## [1] Estimate = 105
## [1] Estimated Standard Error = 1.8974
## [1] Critical Value = 1.6849
## [1] 95% CI = [101.1622,108.8378]
## [1] Test statistic = 2.6352
## [1] Degrees of Freedom = 39
## [1] Pvalue = 0.006
## [1] Decision: reject H0
## [1] =====
```

- a.) State the null and alternative hypotheses

The null hypothesis is that the mean growth rate of venus fly traps is 100mm per year.

$$H_0 : \mu_0 = 100$$

The alternative hypothesis is that the growth rate is higher than 100mm per year

$$H_A : \mu > 100$$

- b.) Compute the critical value for this test

The critical value is $t_{39,0.95} = 1.68$

- c.) Compute the test statistic t_{obs}

$$t_{obs} = \frac{105 - 100}{12/\sqrt{40}} \approx 2.64$$

- d.) Compute the p -value

The p -value is given by

$$P(t > t_{obs} | H_0 \text{ true}) = 1 - P(t \leq t_{obs} | H_0 \text{ true}) = 1 - 0.994 = 0.006$$

- e.) Is there enough evidence to conclude that the current growth rate of Venus fly traps in the greenhouse is higher than the historical value of 100 millimeters per year?

At the 5% significance level, we reject the null hypothesis and conclude that the current growth rate is significantly higher than 100 mm per year.

References

- Agarwal, Ashok, Aditi Mulgund, Alaa Hamada, and Michelle Renee Chyatte. 2015. "A Unique View on Male Infertility Around the Globe." *Reproductive Biology and Endocrinology* 13: 1–9.
- Agresti, Alan, and Christine Franklin. 2007. "The Art and Science of Learning from Data." *Upper Saddle River, New Jersey* 88.
- Alzahrani, Talal, Marwan F Alhazmi, Ahmed N Alharbi, Feras T AlAhmadi, Amer N Alhubayshi, and Bader A Alzahrani. 2023. "The Prevalence of Electronic Cigarette Use Among College Students of Taibah University and Symptoms of Cardiovascular Disease." *Journal of the Saudi Heart Association* 35 (2): 163.
- de Vries, Andrie. 2023. *Surveydata: Tools to Work with Survey Data*. <https://CRAN.R-project.org/package=surveydata>.
- Géron, Aurélien. 2022. *Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow*. " O'Reilly Media, Inc."
- Sharlip, Ira D, Jonathan P Jarow, Arnold M Belker, Larry I Lipshultz, Mark Sigman, Anthony J Thomas, Peter N Schlegel, et al. 2002. "Best Practice Policies for Male Infertility." *Fertility and Sterility* 77 (5): 873–82.
- Zhang, Chenming, Jianshe Chen, Sicheng Ma, Zixue Sun, and Zulong Wang. 2022. "Microplastics May Be a Significant Cause of Male Infertility." *American Journal of Men's Health* 16 (3): 15579883221096549.